

Acknowledgments

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Chapter 1

Introduction

Optimal stopping theory [CRS72, Fer08] studies the fundamental tradeoff between exploration (gathering additional information) and exploitation (taking timely action). Given a sequence of random variables observed at consecutive time steps, an online decision-maker (“gambler”) must determine the point at which it is optimal to stop observing and commit to an action. Optimal stopping theory provides a model to study decision-making under uncertainty, and finds applications in problems such as economics [Luc17] and mechanism design [CHMS10]. A motivating example is the following hiring problem:

You must select a single candidate from a set of n applicants, each with a known prior distribution of quality. Candidates are interviewed sequentially, and their true quality is revealed only at the time of interview. After each interview, you either choose to hire the candidate, or irrevocably reject them. The goal is to maximize the expected quality of the person that is ultimately hired.

This scenario illustrates the inherent tension within optimal stopping: the gambler benefits from interviewing more candidates, since there is no explicit cost to interviewing more; however, there is the risk of missing out on a reasonable candidate in favor of an inferior one [Wei16]. The structure of the distributions, characterized by metrics such as expected value, areas under the CDF curve, etc. are essential to determining the optimal stopping rule.

The prophet inequality, first introduced by Krengel and Sucheston ¹ [KS78, KS77], is an elegant result that compares the performance of a gambler to that of an omniscient “prophet” who knows the entire sequence of realizations in advance. The prophet-gambler ratio thus becomes a measure of this informational asymmetry and yields general competitive ratios

¹and in a private communication by Garling

for the performance of any online strategy. When the values are nonnegative and independent, the **gambler can still achieve at least half the expected reward of a prophet**. Samuel-Cahn [SC84] later showed that this $1/2$ -approximation can be obtained by utilizing a simple threshold strategy: the gambler fixes a threshold T determined from the distributions in advance, and accepts the first observation that exceeds T .

Other classes of prophet inequalities [HK92] include those where the variables are restricted to be independent and identically distributed (*iid*) [HK81, HK82], random-order [EHL17], free-order [Hil83, BGL⁺21, CHMS10], negatively dependent [RSC87], constrained-order [ADK21], etc. A recurring insight across many problems is the existence of threshold rules, suggesting that thresholds capture essential analytical properties of the underlying distributions. Understanding how these thresholds behave, how they depend on distributional and/or order assumptions, and how they generalize beyond the *iid* setting are the primary motivations for the thesis.

1.1 Aim(s)

In the order-selection prophet inequality setting with independent but non-identically distributed random variables, how does the competitive ratio compare to the classical iid case? Does the ratio degrade as the number of distinct distributions increases?

This thesis is motivated by a tension in the literature. In the *iid* case, Hill and Kertz [HK82] show that the worst-case prophet-gambler ratio converges to approximately 0.745. However, many real-world applications are inherently non-*iid*: a sample instance is that of advertising auctions [Haj10] by internet search companies. These can be defined in a prophet inequality setting where businesses submit bids for keywords in advance and user queries — drawn from known but non-*iid* distributions — arrive online — the problem for the search company is to allocate ad spaces to maximize expected revenue.

Order selection (or free-order) is the setting where the gambler chooses the order in which items arrive before the game begins, exploiting distributional knowledge to sequence favorable items. This breaks the $1/2$ -adversarial barrier and achieves a strictly better worst-case competitive ratio than random order; Bubna and Chiplunkar [BC23] establish the current competitive ratio of 0.7258 in the non-*iid* setting. Our central question, which is open, is to characterize the tight lower bound for non-*iid* order selection.

Chapter 2

Background and Related Work

2.1 Origins in Optimal Stopping Theory

The study of prophet inequalities originates in the broader theory of optimal stopping. One of the earliest formalizations of this framework is found in the monograph *Great Expectations* [CRS72] by Chow, Robbins and Siegmund. Optimal stopping problems, in their words, are “the effect of a gambler’s ¹ fortune of various possible systems for deciding when to stop playing a sequence of games”. A central theme in this literature is to quantify the loss due to limited information. If a hypothetical prophet with access to the entire future chooses the maximum value amongst all observed random variables, what fraction of the prophet’s payoff ² can a gambler guarantee? The first formal prophet inequalities appear in the papers by Krengel and Sucheston (and a noted private communication via Garling) [KS77, KS78]. They introduced the term “prophet inequality” and established comparisons between the supremum $\sup_i X_i$ and optimal stopping values for independent random variables under the assumption of adversarial order (the order of viewing items is deterministic and the gambler cannot influence it).

2.2 The Classical 1/2-Prophet Inequality

Setup. Consider³ the following game with n stages: at stage i , you are offered a prize $\pi_i \geq 0$, drawn from a distribution G_i . The distributions are independent and known in advance. The value of π_i is only known at stage i . After seeing π_i , you can either accept the prize and end the game, or discard the prize and proceed to the next stage.

¹In computer science terms: the optimal rewards obtained in online decision problems

²optimal offline reward

³Presented as outlined in [Rou16, SC84].

Theorem 2.1. *For every sequence $G_1 \dots G_n$ of independent distributions, there is a strategy that guarantees expected reward $\frac{1}{2}E_\pi[\max_i \pi_i]$. In fact, there is such a threshold strategy t , which accepts prize i if and only if $\pi_i \geq t$.*

Proof. Let $z^+ = \max(z, 0)$. Fix a threshold t and consider the strategy that accepts the first prize exceeding t . Let

$$q(t) = \Pr(\pi_i < t \ \forall i)$$

be the probability that no prize is accepted. To lower bound the gambler's expected payoff, define

$$Y := t \cdot \mathbf{1}\{\exists i : \pi_i \geq t\} + \sum_{i=1}^n (\pi_i - t) \cdot \mathbf{1}\{\pi_i \geq t, \pi_j < t \ \forall j \neq i\}.$$

When at least one prize exceeds t the gambler collects at least t ; when two or more exceed t , Y credits only t . Thus $Y \leq \text{payoff}$ pointwise, so

$$\mathbb{E}[\text{payoff}] \geq t(1 - q(t)) + \sum_{i=1}^n \mathbb{E}[\pi_i - t \mid \pi_i \geq t, \pi_j < t \ \forall j \neq i] \cdot \Pr(\pi_i \geq t, \pi_j < t \ \forall j \neq i).$$

By independence, we factor the joint probability and drop the conditioning:

$$\mathbb{E}[\text{payoff}] \geq t(1 - q(t)) + \sum_{i=1}^n \mathbb{E}[\pi_i - t \mid \pi_i \geq t] \cdot \Pr(\pi_i \geq t) \cdot \Pr(\pi_j < t \ \forall j \neq i).$$

$\Pr(\pi_j < t \ \forall j \neq i) \geq \Pr(\pi_j < t \ \forall j) = q(t)$ and $\mathbb{E}[(\pi_i - t)^+] = \mathbb{E}[\pi_i - t \mid \pi_i \geq t] \cdot \Pr(\pi_i \geq t)$, we obtain

$$\mathbb{E}[\text{payoff}] \geq t(1 - q(t)) + q(t) \sum_{i=1}^n \mathbb{E}[(\pi_i - t)^+]. \quad (2.1)$$

To upper bound the prophet, we add and subtract t :

$$\mathbb{E}\left[\max_i \pi_i\right] = \mathbb{E}\left[t + \max_i (\pi_i - t)\right] \leq t + \mathbb{E}\left[\max_i (\pi_i - t)^+\right] \leq t + \sum_{i=1}^n \mathbb{E}[(\pi_i - t)^+],$$

where the last step uses $\max_i x_i \leq \sum_i x_i$ for nonneg x_i . Rearranging,

$$\sum_{i=1}^n \mathbb{E}[(\pi_i - t)^+] \geq \mathbb{E}\left[\max_i \pi_i\right] - t.$$

Substituting into (2.1),

$$\mathbb{E}[\text{payoff}] \geq t(1 - 2q(t)) + q(t) \mathbb{E}\left[\max_i \pi_i\right].$$

Since $q(t)$ is non-decreasing with $q(0) = 0$ and $q(t) \rightarrow 1$, we may choose t^* such that $q(t^*) = \frac{1}{2}$. This yields

$$\mathbb{E}[\text{payoff}] \geq \frac{1}{2} \mathbb{E} \left[\max_i \pi_i \right]. \quad \square$$

2.2.1 Tight case

It is easy to see that the 2-factor guarantee is tight [KS77, Luc17] in the adversarial order setting. Consider two items arriving in fixed order.

- The first item has a deterministic reward of 1.
- The second item has reward $1/\epsilon$ with probability ϵ , and 0 otherwise, where $\epsilon > 0$ is arbitrarily small.

The prophet obtains the maximum of the two rewards.

$$\mathbb{E}[\text{OPT}] = \epsilon \cdot \frac{1}{\epsilon} + (1 - \epsilon) \cdot 1 = 2 - \epsilon.$$

In contrast, the gambler must decide whether to take the first item without knowing the realization of the second. If the gambler rejects the first item, its expected reward is only $\epsilon \cdot \frac{1}{\epsilon} = 1$ and if it accepts the first item, it also obtains reward 1. Hence no strategy yields expected reward greater than 1. Therefore the ratio between the prophet's expected reward and the gambler's best possible expected reward approaches 2 as $\epsilon \rightarrow 0$, showing that the 2-approximation barrier is tight.

2.3 Special cases

In this section, we outline some known results of different cases of the prophet inequality from the literature:

2.3.1 IID distributions

Hill and Kertz's 1982 work [HK82] is fundamental to our understanding of the *iid* case. They gave a $1 - 1/e = 0.632$ -competitive algorithm, while showing a hardness bound of 0.745, of which Correa *et al.* [CFH⁺21] later provide a complete characterization. In Hill and Kertz's setting, if T_n denotes the set of stopping rules for the random variables $X_1, \dots, X_n$, then the gambler's optimal value after j stages is defined as $V_j(X) = \sup_{t \in T_j} \mathbb{E}[X_t]$ whereas the prophet's value is given by $E_j(X) = \mathbb{E}[\max(X_1, \dots, X_j)]$.

Structural Insights and Threshold Recursions

A central contribution of Hill and Kertz is the characterization of the optimal stopping values $V_j(X)$ in the iid setting. From Lemma 2.1 (special case of [CRS72] — page 50), using the principle of backwards induction, we have that

$$V_j(X) = \mathbb{E}[\max(X, V_{j-1}(X))].$$

Lemmas 2.4 and 2.8 together provide a two-step reduction.

Lemma 2.4 applies to X taking values in $[0, 1]$: there exists a simple random variable Y supported exactly on $\{0, V_1(X), \dots, V_{n-1}(X), 1\}$

- gambler values are preserved: $V_j(Y) = V_j(X) \ \forall j \in \{1, \dots, n\}$
- prophet values weakly increase: $E_n(Y) \geq E_n(X)$.

Lemma 2.8 handles the unbounded case: for any non-negative unbounded X , there exists a non-negative *bounded* random variable Y satisfying both $V_j(Y) = V_j(X) \ \forall j \in \{1, \dots, n\}$ and $E_n(Y) > E_n(X)$. Together, these results imply that without loss of generality one may restrict attention to simple random variables supported at the gambler's threshold values, and that worst-case distributions for prophet inequalities take precisely this extremal form.

Multiplicative Comparison: Constants a_n

Hill and Kertz prove the following sharpened inequality for iid nonnegative X :

$$E_n(X) \leq a_n V_n(X),$$

where the constant sequence $\{a_n\}$ is the best possible factor for iid distributions, strictly improving upon the factor 2 in the general non-iid case. Some properties of this sequence include:

$$\begin{aligned} 1.1 &< a_n < 1.6, \\ a_2 &\approx 1.171, \quad a_{100} \approx 1.337, \quad a_{10,000} \approx 1.341. \end{aligned}$$

The constants a_n are defined implicitly via an analytic recursion originating in the optimal threshold sequence $V_1(X), \dots, V_n(X)$. A family of functions $\eta_{j,n}$, constructed from the operator

$$\phi_n(w, x) = \frac{n}{n-1} w^{(n-1)/n} + \frac{x}{n-1},$$

is iterated to obtain

$$G_n(\alpha) = \eta_{n-1,n}(\alpha).$$

α_n is defined as the unique solution of $G_n(\alpha_n) = 1$. Hill and Kertz prove that G_n is continuous, non-negative, strictly increasing, and concave, implying the existence and uniqueness of α_n , and set $a_n = 1 + \alpha_n$.

Consequences and Asymptotic Behavior

We are particularly interested in the competitive ratio

$$R_n(X) = \frac{E_n(X)}{V_n(X)}.$$

Probabilistically, $R_n(X)$ is the odds which must be given to a gambler playing against a prophet in order to make the game fair. Hill and Kertz show that for any *fixed* distribution X ,

$$\lim_{n \rightarrow \infty} R_n(X) = 1,$$

so the prophet's advantage vanishes as the horizon grows for any fixed marginal distribution. However, the universal worst-case ratio $\sup_X R_n(X) = a_n$ remains bounded away from 1 for all n , since $a_n > 1.1$. Hill and Kertz conjecture that $\{a_n\}$ and $\{b_n\}$ are monotonically increasing and hence convergent, but were unable to establish this or identify the limiting values.

2.3.2 Random order

This refers to the setting where the gambler sees the items in a uniformly chosen random order. It is also known as the prophet secretary problem [CFH⁺19], and was first studied by Esfandiari *et al.* [EHL17]. Their algorithm defines a non-increasing sequence of n thresholds that are only dependent on the distribution of the maximum of the random variables to find a bound of $1 - 1/e$. Correa *et al.* later give an algorithm that increases the bound to 0.669, and also showed a hardness of $\sqrt{3} - 1 \approx 0.732$. Bubna and Chiplunkar [BC23] improved the hardness bound to 0.7254 for the order-aware variant of the problem, in which the gambler knows the arrival order in advance. For standard random order, Giambartolomei *et al.* [GMTS24] subsequently established a hardness of 0.7235, via a rigorous analytic proof, which also yields an independent separation of random order from order selection.

2.3.3 Order selection

Order selection/free-order refers to the setting where the gambler is allowed to select the order of arrival of items. Bubna and Chiplunkar [BC23] build upon Peng and Tang’s algorithm [PT22] to provide a 0.7258-competitive algorithm. This first established a strict separation between the order selection and random order settings.

2.3.4 Hardness

Agrawal, Sethuraman, and Zhang [ASZ20] show that getting the optimal ordering is NP-hard even for three-point support distributions.

Chapter 3

Worst-Case Competitive Ratio for Two Non-*iid* Distributions

We establish a tight worst-case competitive ratio of $4/5$ for the free-order prophet inequality with two independent non-identically distributed random variables. The proof proceeds in two stages.

1. **Reduction to three-point distributions.** Using a mean-preserving spread rooted in the Hill–Kertz framework, we show that any pair of continuous distributions can be replaced by three-point distributions **without improving** the competitive ratio: the prophet’s value weakly increases while the gambler’s value is exactly preserved.
2. **Optimization over three-point distributions.** Within this family, a mass-shifting argument reduces further to the case in which both possible orderings yield the same gambler value. An explicit calculation over this subfamily then yields the tight bound of $4/5$.

Lemma 3.1 (Truncation/Boundedness Lemma). *Let A and B be independent integrable random variables with finite means. For $M > 0$, define $A_M = \min(A, M)$ and $B_M = \min(B, M)$. Then as $M \rightarrow \infty$:*

$$\left| \text{OPT}_{\text{off}}(A, B) - \text{OPT}_{\text{off}}(A_M, B_M) \right| \rightarrow 0, \quad \left| \text{OPT}_{\text{on}}(A, B) - \text{OPT}_{\text{on}}(A_M, B_M) \right| \rightarrow 0.$$

Proof. The payoff difference is bounded above by $\mathbb{E}[A \cdot \mathbf{1}_{A > M}] + \mathbb{E}[B \cdot \mathbf{1}_{B > M}]$, which tends to 0 as $M \rightarrow \infty$ by the dominated convergence theorem since A and B are integrable. $\square$

3.1 Reduction to Three-Point Distributions

Let A and B be independent integrable random variables with finite means $\mu_A = \mathbb{E}[A]$ and $\mu_B = \mathbb{E}[B]$, and let $M > 0$ be an upper bound on their support (by Lemma 3.1, it suffices to prove the result for bounded random variables). We construct three-point discretizations

$$A' \in \{0, \mu_B, M\} \quad \text{and} \quad B' \in \{0, \mu_A, M\}$$

that preserve the respective means.

3.1.1 Construction

For each realization x of A , we define the conditional distribution of A' is defined by linear interpolation to the nearest endpoints of each sub-interval:

$$\begin{aligned} \text{For } x \in [0, \mu_B] : \quad & P(A' = 0 \mid X = x) = \frac{\mu_B - x}{\mu_B}, \quad P(A' = \mu_B \mid X = x) = \frac{x}{\mu_B}; \\ \text{for } x \in [\mu_B, M] : \quad & P(A' = \mu_B \mid X = x) = \frac{M - x}{M - \mu_B}, \quad P(A' = M \mid X = x) = \frac{x - \mu_B}{M - \mu_B}. \end{aligned}$$

The discretization B' is defined symmetrically, with threshold μ_A in place of μ_B :

$$\begin{aligned} \text{For } x \in [0, \mu_A] : \quad & P(B' = 0 \mid X = x) = \frac{\mu_A - x}{\mu_A}, \quad P(B' = \mu_A \mid X = x) = \frac{x}{\mu_A}; \\ \text{for } x \in [\mu_A, M] : \quad & P(B' = \mu_A \mid X = x) = \frac{M - x}{M - \mu_A}, \quad P(B' = M \mid X = x) = \frac{x - \mu_A}{M - \mu_A}. \end{aligned}$$

3.1.2 Mean Preservation

Note that on each interval the map $x \mapsto \mathbb{E}[A' \mid X = x]$ recovers x exactly:

$$\text{On } [0, \mu_B] : 0 \cdot \frac{\mu_B - x}{\mu_B} + \mu_B \cdot \frac{x}{\mu_B} = x; \quad \text{on } [\mu_B, M] : \mu_B \cdot \frac{M - x}{M - \mu_B} + M \cdot \frac{x - \mu_B}{M - \mu_B} = x.$$

and hence $\mathbb{E}[A'] = \mathbb{E}[\mathbb{E}[A' \mid X]] = \mathbb{E}[A] = \mu_A$. An identical argument gives $\mathbb{E}[B'] = \mu_B$.

3.1.3 Discretization Worsens the Competitive Ratio

Let G_{AB} and G_{BA} denote the gambler's expected reward under orderings $A \rightarrow B$ and $B \rightarrow A$ respectively.

Lemma 3.2. *The three-point discretizations A' and B' satisfy:*

i) **Prophet value weakly increases:** $\mathbb{E}[\max(A, B)] \leq \mathbb{E}[\max(A', B')]$.

ii) **Gambler value exactly preserved:** $G_{A'B'} = G_{AB}$.

Consequently, the competitive ratio is non-increasing under this discretization.

Proof. (i). We apply Lemma 2.2 of [HK81] (equivalently Lemma 2.3 of [HK82]) in two steps for A , using Definition 2.2 of [HK81] on each sub-interval separately. First, spread A on $[0, \mu_B]$ to obtain $A_1 = A_0^{\mu_B}$; by Lemma 2.2 of [HK81], $\mathbb{E}[\max(A, B)] \leq \mathbb{E}[\max(A_1, B)]$. Second, spread A_1 on $[\mu_B, M]$ to obtain $A' = (A_1)_{\mu_B}^M$; applying Lemma 2.2 of [HK81] again,

$$\mathbb{E}[\max(A_1, B)] \leq \mathbb{E}[\max(A', B)].$$

The result of both steps is $A' \in \{0, \mu_B, M\}$. Applying the identical two-step argument to spread $B \rightarrow B' \in \{0, \mu_A, M\}$ gives

$$\mathbb{E}[\max(A', B)] \leq \mathbb{E}[\max(A', B')].$$

Chaining the three inequalities yields $\mathbb{E}[\max(A, B)] \leq \mathbb{E}[\max(A', B')]$.

(ii). By Lemma 2.1 of [HK81], the gambler's optimal threshold for A in order AB is $V(B) = \mathbb{E}[B] = \mu_B$, so the gambler's expected reward is

$$G_{AB} = \mathbb{E}[(A - \mu_B)^+] + \mu_B.$$

It suffices to show $\mathbb{E}[(A' - \mu_B)^+] = \mathbb{E}[(A - \mu_B)^+]$. The function $f(x) = (x - \mu_B)^+$ is identically zero on $[0, \mu_B]$ and linear (equal to $x - \mu_B$) on $[\mu_B, M]$. $A \mapsto A'$ (Definition 2.2 of [HK81]) is preserves the mean within each sub-interval, so by linearity of f on each piece and the law of total expectation:

$$\mathbb{E}[f(A') \mid A = x] = f(\mathbb{E}[A' \mid A = x]) = f(x) \quad \text{for all } x,$$

and therefore $\mathbb{E}[(A' - \mu_B)^+] = \mathbb{E}[(A - \mu_B)^+]$, giving $G_{A'B} = G_{AB}$. For the second step, the threshold for B in order BA' is $\mathbb{E}[A'] = \mathbb{E}[A] = \mu_A$ (mean preserved by the spread). Using an identical argument, we obtain $G_{A'B'} = G_{A'B} = G_{AB}$. $\square$

Corollary 3.3. *Hence, it suffices to prove worst-case bounds for pairs of three-point distributions $A \in \{0, \mu_B, M\}$ and $B \in \{0, \mu_A, M\}$.*

3.2 Worst-Case Ratio over Three-Point Distributions

3.2.1 Parametrization

Definition 3.4 (Probability parameters). Let

$$P(A = 0) = \alpha_0, \quad P(A = \mu_B) = \alpha_1, \quad P(A = M) = \alpha_2,$$

$$P(B = 0) = \beta_0, \quad P(B = \mu_A) = \beta_1, \quad P(B = M) = \beta_2,$$

where $\alpha_0 + \alpha_1 + \alpha_2 = 1$ and $\beta_0 + \beta_1 + \beta_2 = 1$. The mean constraints read

$$\mu_A = \alpha_1 \mu_B + \alpha_2 M, \quad \mu_B = \beta_1 \mu_A + \beta_2 M.$$

The CDFs of A and B are step functions with jumps at 0 , μ_B (resp. μ_A), and M , as shown in Figure 3.1.

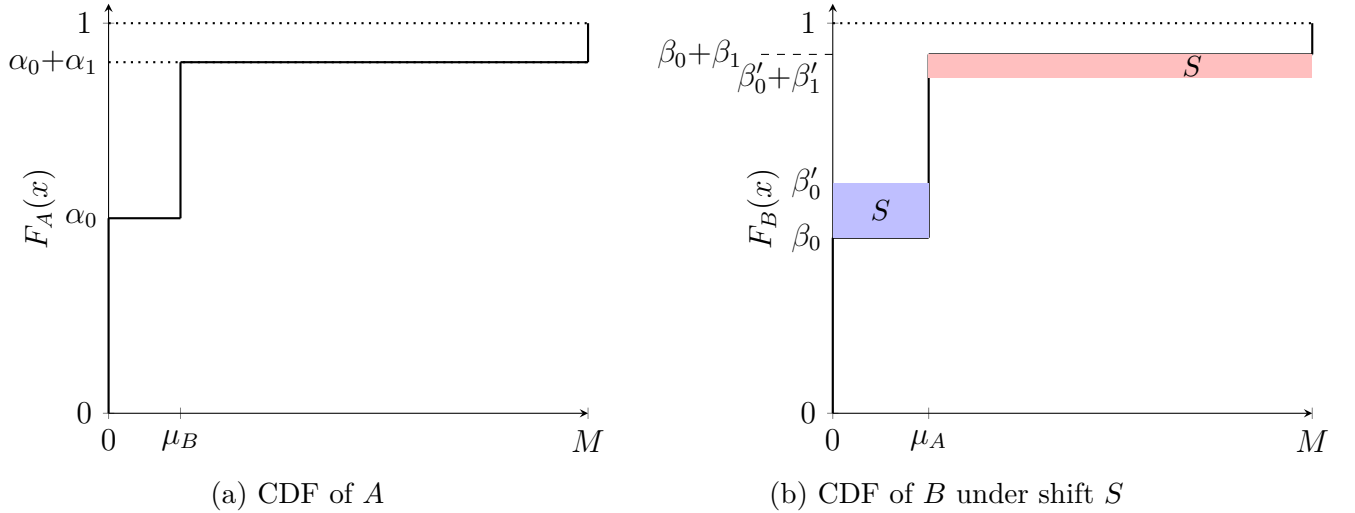


Figure 3.1: Three-point CDFs of A and B .

3.2.2 Probability Mass Shifting

We introduce a probability mass shift that spreads the mass of B .

Definition 3.5 (Shifted distribution of B). For a shift parameter $S \geq 0$, define

$$\beta'_0 = \beta_0 + \frac{S}{\mu_A}, \quad \beta'_1 = \beta_1 - \frac{S}{\mu_A} - \frac{S}{M - \mu_A}, \quad \beta'_2 = \beta_2 + \frac{S}{M - \mu_A}.$$

We can verify that $\beta'_0 + \beta'_1 + \beta'_2 = 1$ and $\beta'_1 \mu_A + \beta'_2 M = \mu_B$ (the mean is unchanged).

Lemma 3.6 (Feasibility). *The shifted distribution is valid if and only if $S \leq \beta_1 \mu_A (M - \mu_A)/M$. In the limit $M \rightarrow \infty$ this simplifies to $S \leq \beta_1 \mu_A$.*

Proof. To obtain a valid distribution we require $\beta'_1 \geq 0$, i.e. $\beta_1 \geq S/\mu_A + S/(M - \mu_A) = SM/(\mu_A(M - \mu_A))$. $\square$

Theorem 3.7 (Shift increases the prophet's value). *For $S > 0$,*

$$\mathbb{E}[\max(A, B')] - \mathbb{E}[\max(A, B)] = \frac{S \mu_B \alpha_1}{\mu_A} \geq 0,$$

with strict inequality whenever $\alpha_1 > 0$.

Proof. Assume WLOG $\mu_B \leq \mu_A$. The prophet's payoff equals

$$\mathbb{E}[\max(A, B)] = \mu_B \alpha_1 \beta_0 + \mu_A (\alpha_0 + \alpha_1) \beta_1 + M [1 - (1 - \alpha_2)(1 - \beta_2)]. \quad (3.1)$$

Computing the change under the shift:

$$\begin{aligned} \mathbb{E}[\max(A, B')] - \mathbb{E}[\max(A, B)] &= (\beta'_0 - \beta_0) \mu_B \alpha_1 + (\beta'_1 - \beta_1) (\alpha_0 + \alpha_1) \mu_A + (\beta'_2 - \beta_2) (1 - \alpha_2) M \\ &= \frac{S}{\mu_A} \mu_B \alpha_1 - \left(\frac{S}{\mu_A} + \frac{S}{M - \mu_A} \right) (1 - \alpha_2) \mu_A + \frac{S}{M - \mu_A} (1 - \alpha_2) M. \end{aligned}$$

Using $\alpha_0 + \alpha_1 = 1 - \alpha_2$ and collecting terms:

$$\begin{aligned} &= \frac{S \mu_B \alpha_1}{\mu_A} - S(1 - \alpha_2) - \frac{S(1 - \alpha_2) \mu_A}{M - \mu_A} + \frac{S(1 - \alpha_2) M}{M - \mu_A} \\ &= \frac{S \mu_B \alpha_1}{\mu_A} - S(1 - \alpha_2) + S(1 - \alpha_2) \\ &= \frac{S \mu_B \alpha_1}{\mu_A}. \end{aligned} \quad \square$$

3.2.3 Reduction to the Equal-Order Case

The free-order gambler chooses the better ordering, so the gambler's value is $\max(G_{AB}, G_{BA})$.

Theorem 3.8 (Worst case has equal orderings). *For every pair (A, B) of three-point distributions with $G_{AB} \geq G_{BA}$, there exists a distribution B^* such that:*

1. $G_{AB^*} = G_{B^*A}$, and
2. the competitive ratio with (A, B^*) is at most that with (A, B) .

Proof. Apply the mass shift of Definition 3.5, increasing S continuously from 0 to $S_{\max} = \beta_1 \mathbb{E}[A]$. At $S = 0$, we have $G_{AB} \geq G_{BA}$ by assumption. At $S = S_{\max}$, $\beta'_1 = 0$, so B' is supported on $\{0, M\}$ only. In this case it is weakly optimal to inspect B' first: doing so either yields M (the maximum possible value, so inspection stops) or reveals $B' = 0$, after which the player knows not to take B' and proceeds to inspect A . Hence inspecting B' first is never worse than inspecting A first, giving $G_{B'A} \geq G_{AB'}$. Since $G_{AB'}$ and $G_{B'A}$ vary continuously in S , by the intermediate value theorem there exists $S^* \in (0, S_{\max})$ at which $G_{AB'} = G_{B'A}$. Set $B^* = B'|_{S=S^*}$, establishing (1).

By Theorem 3.7, the prophet's value $\mathbb{E}[\max(A, B')]$ is strictly increasing in S . Meanwhile, the gambler's value is unchanged: since the shift preserves $\mathbb{E}[B'] = \mathbb{E}[B] = \mu_B$, the gambler's threshold for A in order AB' remains μ_B for all S , so

$$G_{AB'} = \mathbb{E}[(A - \mu_B)^+] + \mu_B = G_{AB} \quad \text{for all } S \in [0, S^*].$$

Hence the gambler's value $\max(G_{AB'}, G_{B'A}) = G_{AB'} = G_{AB}$ is constant on $[0, S^*]$, while the prophet's value increases. Therefore the competitive ratio at (A, B^*) is at most that at (A, B) , establishing (2). $\square$

3.3 Computing the Tight Ratio

Having proven that the equal-ordering case 3.8 yields the worst-case ratio, we derive the tight bound under that assumption:

Definition 3.9 (Continuation probabilities). Write the means as

$$\mu_A = P_A \mu_B + C_A, \quad \mu_B = P_B \mu_A + C_B,$$

where $P_A = \alpha_1 \in [0, 1]$ is the probability A takes the value μ_B , $C_A = \alpha_2 M$ is the expected contribution from A 's large point mass, and P_B, C_B are defined symmetrically for B .

Using Definition 3.9, the gambler's expected rewards are:

$$G_{AB} = C_A + \mu_B = C_A + C_B + P_B \mu_A, \tag{3.2}$$

$$G_{BA} = C_B + \mu_A = C_A + C_B + P_A \mu_B. \tag{3.3}$$

Lemma 3.10 (Symmetry condition). $G_{AB} = G_{BA}$ if and only if $P_A \mu_B = P_B \mu_A$.

Proof. Subtracting (3.3) from (3.2) gives $G_{AB} - G_{BA} = P_B \mu_A - P_A \mu_B$. $\square$

Lemma 3.11 (Normalization). *Assume $\mu_B \leq \mu_A$ and normalize $\mu_A = 1$. Then the symmetry condition $P_A \mu_B = P_B$ holds, and the common gambler value is*

$$G^* := G_{AB} = G_{BA} = 1 + (1 - P_A) \mu_B.$$

Proof. With $\mu_A = 1$, the symmetry condition gives $P_B = P_A \mu_B$. Then $G_{AB} = C_A + \mu_B = (\mu_A - P_A \mu_B) + \mu_B = 1 + (1 - P_A) \mu_B$. $\square$

Theorem 3.12 (Worst-case competitive ratio). *Under the symmetric equal-ordering condition ($G_{AB} = G_{BA} = G^*$),*

$$\frac{G^*}{\mathbb{E}[\max(A, B)]} \geq \frac{4}{5},$$

with equality in the limit $\mu_B \rightarrow P_B \mu_A$ and $P_B \rightarrow \frac{1}{2}$.

Proof. Normalize $\mu_A = 1$. Under the symmetry condition $P_A \mu_B = P_B \mu_A$, we have $k := P_A \mu_B = P_B$. Substituting into (3.2)–(3.3),

$$G^* = 1 + \mu_B - P_B.$$

Using (3.1) in the limit $M \rightarrow \infty$ with $\alpha_2 M = 1 - P_B$ and $\beta_2 M = \mu_B - P_B$ held fixed:

$$\begin{aligned} \mathbb{E}[\max(A, B)] &= \mu_B \cdot \frac{P_B}{\mu_B} \cdot (1 - P_B) + 1 \cdot 1 \cdot P_B + (1 - P_B) + (\mu_B - P_B) \\ &= P_B(1 - P_B) + P_B + 1 - P_B + \mu_B - P_B \\ &= 1 + \mu_B - P_B^2. \end{aligned}$$

Since $\mathbb{E}[\max(A, B)] - G^* = P_B(1 - P_B)$,

$$\frac{\mathbb{E}[\max(A, B)]}{G^*} = 1 + \frac{P_B(1 - P_B)}{1 + \mu_B - P_B}.$$

The feasibility constraint $P_A = P_B / \mu_B \leq 1$ requires $\mu_B \geq P_B$, so $1 + \mu_B - P_B \geq 1$. Therefore

$$\frac{P_B(1 - P_B)}{1 + \mu_B - P_B} \leq P_B(1 - P_B) \leq \frac{1}{4},$$

and hence

$$\frac{G^*}{\mathbb{E}[\max(A, B)]} \geq \frac{1}{1 + \frac{1}{4}} = \frac{4}{5}.$$

$\square$

3.3.1 A Tight Example

We exhibit a pair of distributions attaining the bound $4/5$ in the limit. Fix $\varepsilon > 0$ small and define:

Distribution of B .

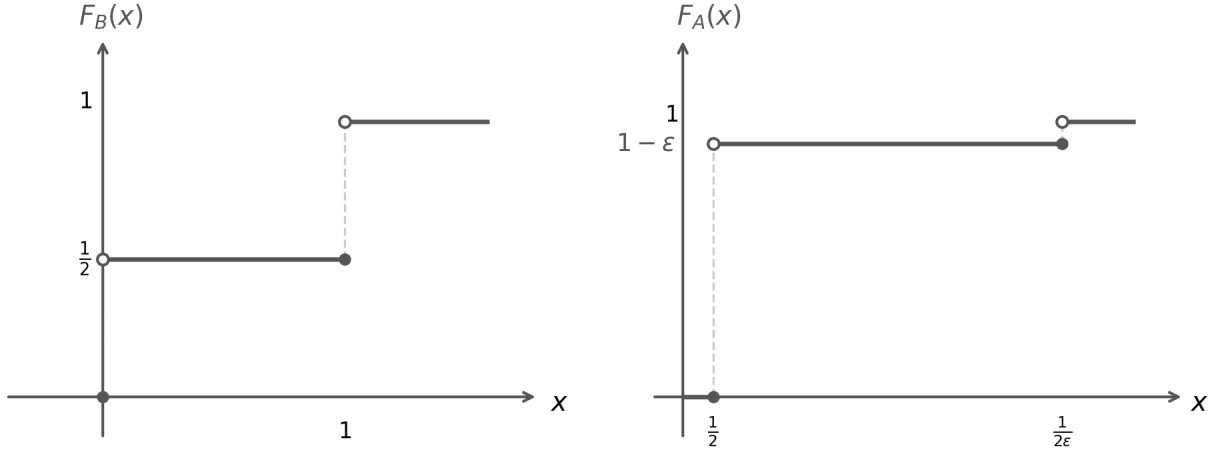
$$B = \begin{cases} 0 & \text{with probability } \frac{1}{2}, \\ 1 & \text{with probability } \frac{1}{2}. \end{cases}$$

Distribution of A .

$$A = \begin{cases} \frac{1}{2} & \text{with probability } 1 - \varepsilon, \\ \frac{1}{2\varepsilon} & \text{with probability } \varepsilon. \end{cases}$$

Then

$$\mathbb{E}[B] = \frac{1}{2}, \quad \mathbb{E}[A] = (1 - \varepsilon) \cdot \frac{1}{2} + \varepsilon \cdot \frac{1}{2\varepsilon} = \frac{1}{2}(1 - \varepsilon) + \frac{1}{2} = 1 - \frac{\varepsilon}{2} \xrightarrow{\varepsilon \rightarrow 0} 1.$$



Worst-case distribution CDFs.

Figure 3.2: CDFs of the worst-case distributions A and B .

Gambler's value. In order $A \rightarrow B$: the gambler's threshold for A is $\mathbb{E}[B] = 1/2$. Both outcomes of A satisfy $A \geq 1/2$, so the gambler always accepts A , obtaining $G_{AB} = \mathbb{E}[A] \rightarrow 1$. In order $B \rightarrow A$: the threshold for B is $\mathbb{E}[A] \rightarrow 1$. Only $B = 1$ clears this threshold, so the gambler accepts $B = 1$ (probability $1/2$) and otherwise accepts A (expected value $\rightarrow 1$); hence $G_{BA} \rightarrow 1/2 \cdot 1 + 1/2 \cdot 1 = 1$.

Prophet's value. The maximum takes values $1/2$, 1 , and $1/(2\varepsilon)$ with probabilities $(1 - \varepsilon) \cdot 1/2$, $(1 - \varepsilon) \cdot 1/2$, and ε respectively:

$$\begin{aligned}\mathbb{E}[\max(A, B)] &= \frac{1}{2\varepsilon} \cdot \varepsilon + 1 \cdot (1 - \varepsilon) \cdot \frac{1}{2} + \frac{1}{2} \cdot (1 - \varepsilon) \cdot \frac{1}{2} \\ &= \frac{1}{2} + \frac{3(1 - \varepsilon)}{4} \xrightarrow{\varepsilon \rightarrow 0} \frac{5}{4}.\end{aligned}$$

Chapter 4

Exploring three non-*iid* distributions

4.1 Three Distributions: Equal Permutations

We now turn to the three-distribution case. Motivated by the two-distribution analysis, we focus first on the **equal-permutations** regime: the candidate worst-case instances for which the gambler obtains the same value from every possible ordering. This setup is based on the work done in an untitled manuscript by Beyhaghi [Bey19].

In the two-distribution problem, the worst-case construction occurs when the two possible orders give the same gambler value. Our working hypothesis is that an analogous phenomenon persists for three distributions, so the equal-permutations case is the natural first place to search for the worst counterexample. This is also supported by the computational results in [Section 4.1.3](#).

Notation. Let X_A, X_B, X_C denote the three independent random variables. For brevity, we write

$$A := \mathbb{E}[X_A], \quad B := \mathbb{E}[X_B], \quad C := \mathbb{E}[X_C].$$

and

$$BC := \text{the gambler's expected value when the remaining order is } (B, C),$$

with analogous meanings for AB , AC , and so on. Under the equal-permutations hypothesis, reversing the order of the final two variables does not change the continuation value, so we may identify $BC = CB$, $AC = CA$, and $AB = BA$.

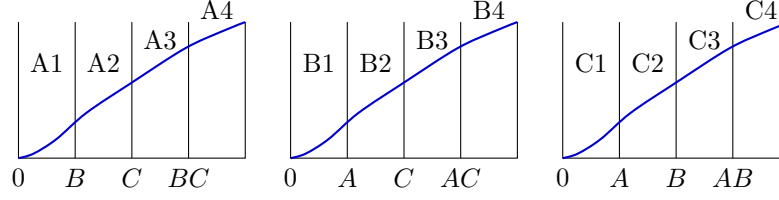


Figure 4.1: CDF of distributions X_A, X_B, X_C . WLOG we assume $A \leq B \leq C$.

4.1.1 Setup and Parameterization

Why these support points? We extend the threshold-support argument of Lemma 3.2 to three distributions via backward induction. Fixing a distribution X_A and enumerating all six orderings of (A, B, C) : by backward induction, the gambler's optimal threshold for X_A in a given ordering equals the continuation value of the remaining subproblem:

Orderings	Position of A	Threshold for X_A
(A, B, C) and (A, C, B)	first	BC
(B, A, C) and (C, A, B)	second	C and B , respectively
(B, C, A) and (C, B, A)	last	0 (must accept)

The set of thresholds that X_A can face across all orderings is therefore $\{0, B, C, BC\}$. By the same mean-preserving spread argument as Lemma 3.2 applied to each consecutive pair of thresholds, any mass of X_A in an open interval (t_i, t_{i+1}) between consecutive thresholds can be spread to the two endpoints without changing the gambler's accept/reject decision in *any* ordering, while weakly increasing the prophet's value. It therefore suffices to consider X_A supported on $\{0, B, C, BC\}$. Symmetric reasoning gives support $\{0, A, C, AC\}$ for X_B and $\{0, A, B, AB\}$ for X_C .

Area parameters. The quantities A_1, A_2, A_3, A_4 are the areas above the CDF of X_A on the intervals that correspond respectively to the intervals $[0, B)$, $[B, C)$, $[C, BC)$, and the terminal point $\{BC\}$. Equivalently, they are the contributions of those intervals to the survival-function integral for X_A . Once these areas are fixed, the associated densities on each interval are obtained by dividing by the interval length, which explains the constraints that appear below. For the offline optimization, an extremal distribution can then be taken to place its mass on interval endpoints, so these area parameters are enough to recover the candidate worst-case distributions.

Lemma 4.1. $AB \leq AC \leq BC$.

Proof. In the equal-permutations setting, $AB = \mathbb{E}[\max(X_A, B)]$, $AC = \mathbb{E}[\max(X_A, C)] = \mathbb{E}[\max(X_C, A)]$, and $BC = \mathbb{E}[\max(X_C, B)]$. Since $B \leq C$,

$$AC - AB = \mathbb{E}[\max(X_A, C) - \max(X_A, B)] \geq 0.$$

Since $A \leq B$,

$$BC - AC = \mathbb{E}[\max(X_C, B) - \max(X_C, A)] \geq 0. \quad \square$$

$$A \leq B \leq C, AB \leq AC \leq BC$$

□

We also normalize the online optimum to equal 1. This is the constraint $BC + A_4 = 1$. Distribution A is therefore parameterized by the areas A_1, A_2, A_3, A_4 over the intervals $[0, B)$, $[B, C)$, $[C, BC)$, and $\{BC\}$. Once A_1, A_2, A_3, A_4, B, C , and BC are chosen, all remaining quantities are determined uniquely:

$$\begin{aligned} AC &= C + A_3 + A_4, & AB &= B + A_2 + A_3 + A_4, \\ B_1 &= A_1, & B_3 &= A_3, & B_4 &= BC - C - A_3, & B_2 &= B - A_1 - BC + C, \\ C_1 &= A_1 + A_2, & C_3 &= A_2 + A_3, & C_4 &= BC - B - A_2 - A_3, & C_2 &= B + C - BC - A_1 - A_2. \end{aligned}$$

The inequalities

$$\frac{A_1}{B} \geq \frac{A_2}{C-B} \geq \frac{A_3}{BC-C}, \quad \frac{B_1}{A} \geq \frac{B_2}{C-A} \geq \frac{B_3}{AC-C}, \quad \frac{C_1}{A} \geq \frac{C_2}{B-A} \geq \frac{C_3}{AB-B}$$

ensure that these piecewise-constant densities are well defined and non-increasing.

4.1.2 Offline Optimum

The prophet's value equals $\int_0^\infty (1 - F_{\max(A,B,C)}(x)) dx$. Because the distributions are supported on finitely many intervals, this reduces to a sum of terms, one per interval of the maximum's support. The formula depends on whether $AB \leq C$ or $C \leq AB$.

Case 1: $C \leq AB$.

$$\begin{aligned}
\text{OPT}_{\text{off}} = & A \left[1 - \left(1 - \frac{A_1}{B} \right) \left(1 - \frac{B_1}{A} \right) \left(1 - \frac{C_1}{A} \right) \right] \\
& + (B - A) \left[1 - \left(1 - \frac{A_1}{B} \right) \left(1 - \frac{B_2}{C - A} \right) \left(1 - \frac{C_2}{B - A} \right) \right] \\
& + (C - B) \left[1 - \left(1 - \frac{A_2}{C - B} \right) \left(1 - \frac{B_2}{C - A} \right) \left(1 - \frac{C_3}{AB - B} \right) \right] \\
& + (AB - C) \left[1 - \left(1 - \frac{A_3}{BC - C} \right) \left(1 - \frac{B_3}{AC - C} \right) \left(1 - \frac{C_3}{AB - B} \right) \right] \\
& + (AC - AB) \left[1 - \left(1 - \frac{A_3}{BC - C} \right) \left(1 - \frac{B_3}{AC - C} \right) \right] \\
& + (BC - AC) \left[1 - \left(1 - \frac{A_3}{BC - C} \right) \right] + A_4 + B_4 + C_4.
\end{aligned}$$

Case 2: $AB \leq C$.

$$\text{OPT}_{\text{off}} = A \left[1 - \left(1 - \frac{A_1}{B} \right) \left(1 - \frac{B_1}{A} \right) \left(1 - \frac{C_1}{A} \right) \right] \quad (4.1)$$

$$+ (B - A) \left[1 - \left(1 - \frac{A_1}{B} \right) \left(1 - \frac{B_2}{C - A} \right) \left(1 - \frac{C_2}{B - A} \right) \right] \quad (4.2)$$

$$+ (AB - B) \left[1 - \left(1 - \frac{A_2}{C - B} \right) \left(1 - \frac{B_2}{C - A} \right) \left(1 - \frac{C_3}{AB - B} \right) \right] \quad (4.3)$$

$$+ (C - AB) \left[1 - \left(1 - \frac{A_2}{C - B} \right) \left(1 - \frac{B_2}{C - A} \right) \right] \quad (4.4)$$

$$+ (AC - C) \left[1 - \left(1 - \frac{A_3}{BC - C} \right) \left(1 - \frac{B_3}{AC - C} \right) \right] \quad (4.5)$$

$$+ (BC - AC) \left[1 - \left(1 - \frac{A_3}{BC - C} \right) \right] \quad (4.6)$$

$$+ A_4 + B_4 + C_4. \quad (4.7)$$

In both cases the optimization problem is to maximize OPT_{off} subject to $A_1, A_2, A_3, A_4, B_4, C_4 \geq 0$, $A_1 + A_2 + A_3 + A_4 \leq B \leq C \leq BC$, $BC + A_4 = 1$, and the density-ratio inequalities.

4.1.3 Computational Experiments

To guide the analysis we solved both optimization problems numerically using `scipy` [VGO⁺20]. We used `scipy.optimize.minimize` with the SLSQP (Sequential Least Squares Programming) method. SLSQP is a gradient-based method that handles nonlinear constraints by

solving a quadratic programming subproblem at each iterate; because it is a local method, we ran 5,000 random restarts per case to alleviate sensitivity to initialization. The nonlinear constraints (ordering, normalization, case condition (case 1 or 2), and all three sets of density-ratio inequalities) were encoded as `NonlinearConstraint` objects. Across all restarts, the best objective value found for both cases converged to ≈ 1.25 .

4.1.4 A Structural Conjecture

The numerical experiments consistently returned solutions with $A_3 \approx 0$ and $A_4 \approx 0$. This motivates the following conjecture, which we verify analytically.

Conjecture 4.2. *The worst-case ratio in the equal-permutations, three-distribution setting is attained (in the limit) by instances satisfying $A_3 = A_4 = 0$.*

We now show that $A_3 = A_4 = 0$ forces $C = BC = 1$ under the normalization $BC + A_4 = 1$, and that these four conditions together imply the worst-case ratio is exactly $\frac{5}{4}$.

Lemma 4.3. *If $A_3 = A_4 = 0$, then $BC = C = 1$*

Proof. Plugging in $A_3 = A_4 = 0$ into the structural definitions, $AC = C$, $BC = 1$, and $AB = B + A_2$. Since $AB \leq AC \leq BC$, we have $AB \leq C$ (Case II). CDF monotonicity of C requires:

$$\frac{C_1}{A} \geq \frac{C_2}{B-A} \geq \frac{C_3}{AB-B}.$$

From the right inequality, plugging in the values of C_2, C_3, AB :

$$\frac{B-A+C-BC}{B-A} \geq \frac{A_2}{B+A_2-B} = 1,$$

which simplifies to:

$$1 + \frac{C-BC}{B-A} \geq 1,$$

and therefore $C \geq BC$. However, since $AB \leq AC = C \leq BC$, we also have $C \leq BC$. Together these yield $C = BC = 1$. $\square$

Proof. From $BC + A_4 = 1$ and $A_4 = 0$, we immediately get $BC = 1$. Then $AC = C + A_3 + A_4 = C + 0 + 0 = C$, and $B_4 = BC - C - A_3 = 1 - C - 0 = 1 - C \geq 0$. The numerical optimizer places all solutions near $C = 1$, consistent with the upper bound $C \leq BC = 1$. We therefore set $C = 1$. $\square$ $\square$

4.1.5 Proof That Conjectured Worst-Case Ratio Is $5/4$

Theorem 4.4. *Under the assumptions of Conjecture 4.2, in the equal-permutations setting with three distributions, the worst-case ratio $\text{OPT}_{\text{off}}/\text{OPT}_{\text{on}}$ is at most $\frac{5}{4}$.*

Proof. By Lemma 4.3 we work under $A_3 = A_4 = 0$ and $C = BC = 1$. Writing $A = A_1 + A_2$, the derived quantities simplify to

$$B_1 = A_1, \quad B_2 = B - A_1, \quad B_3 = B_4 = 0,$$

$$C_1 = A, \quad C_2 = B - A, \quad C_3 = A_2, \quad C_4 = 1 - B - A_2.$$

Since $AB = B + A_2 \leq C = 1$, Case 2 applies. Set $y := C_4 = 1 - B - A_2$ and evaluate each term of (4.1)–(4.7):

- **Term (4.1).** Since $C_1 = A$, the factor $(1 - C_1/A) = 0$, so the term equals A .
- **Term (4.2).** Since $C_2 = B - A$, the factor $(1 - C_2/(B - A)) = 0$, so the term equals $B - A$.
- **Term (4.3).** Since $C_3 = A_2 = AB - B$, the factor $(1 - C_3/(AB - B)) = 0$, so the term equals $AB - B = A_2$.
- **Term (4.4).** The prefactor is $C - AB = 1 - B - A_2 = y$. Using $1 - B - A_2 = y$ in the first fraction and $1 - A - B + A_1 = 1 - B - A_2 = y$ in the numerator of the second:

$$(4.4) = y \left[1 - \frac{y}{1 - B} \cdot \frac{y}{1 - A} \right] = y - \frac{y^3}{(1 - B)(1 - A)}.$$

- **Terms (4.5) and (4.6).** The prefactors are $AC - C = 0$ and $BC - AC = 0$, so both vanish.
- **Term (4.7).** $A_4 + B_4 + C_4 = 0 + 0 + y = y$.

Summing and using $B + A_2 = 1 - y$:

$$\text{OPT}_{\text{off}} = \underbrace{A + (B - A) + A_2}_{=1-y} + y - \frac{y^3}{(1 - B)(1 - A)} + y = 1 + y - \frac{y^3}{(1 - B)(1 - A)}.$$

It remains to bound the denominator. The density-ratio constraint $A_1/B \geq A_2/(1 - B)$ gives $A_1(1 - B) \geq A_2B$, hence $A_1 \geq AB$. Therefore:

$$(1 - B)(1 - A) = 1 - A - B + AB \leq 1 - A - B + A_1 = 1 - B - A_2 = y,$$

so $y^3/(1-B)(1-A) \geq y^2$, giving

$$\text{OPT}_{\text{off}} \leq 1 + y - y^2.$$

The quadratic $f(y) = 1 + y - y^2$ is maximized at $y = \frac{1}{2}$, where $f(\frac{1}{2}) = \frac{5}{4}$. Since $\text{OPT}_{\text{on}} = BC + A_4 = 1$, the ratio is at most $\frac{5}{4}$. $\square$

4.1.6 A Conjectured Tight Example for the Worst-Case Ratio

We now show a matching construction for the possible worst-case bound $\frac{5}{4}$. Fix small $\varepsilon > 0$ and set $p = 1 - 1/\sqrt{2}$. Define three independent random variables:

$$A = \begin{cases} 0 & \text{w.p. } 1-p, \\ 1 & \text{w.p. } p, \end{cases} \quad B = \begin{cases} 0 & \text{w.p. } 1-p, \\ 1 & \text{w.p. } p, \end{cases} \quad C = \begin{cases} \frac{1}{2} & \text{w.p. } 1-\varepsilon, \\ \frac{1}{2\varepsilon} & \text{w.p. } \varepsilon. \end{cases}$$

This instance corresponds to the boundary $y = 1/2$ and $(1-B)(1-A) = y$ in the proof above.

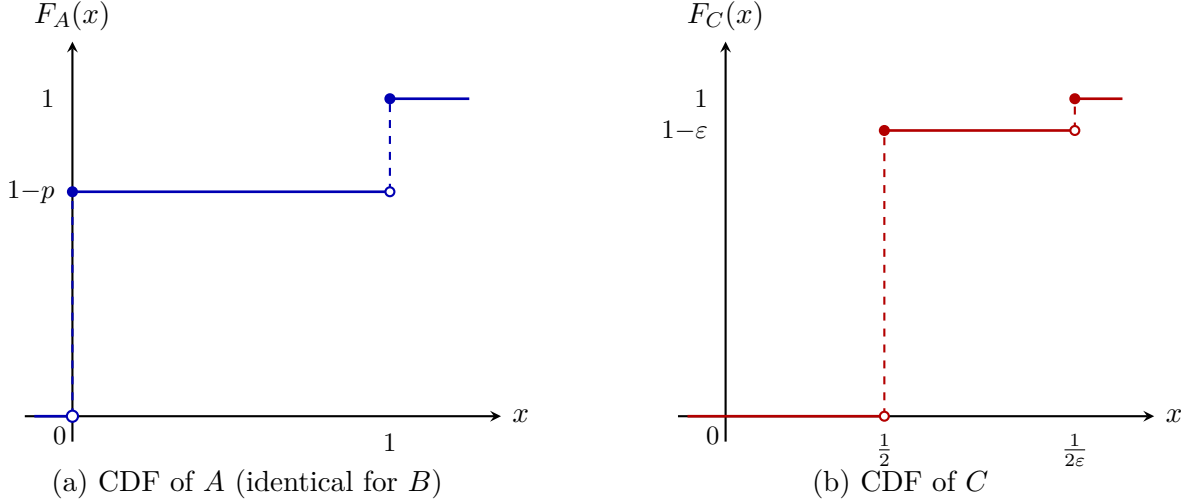


Figure 4.2: CDFs of $A (= B)$ and C . The variable C places most of its mass at $\frac{1}{2}$ but has a rare large outcome at $\frac{1}{2\varepsilon}$.

Gambler's Value The gambler observes A, B, C sequentially in some fixed order and must irrevocably accept or reject each value upon seeing it. We verify by backward induction that each of the following orderings yields gambler's value 1.

- **Order ABC .** At stage 3 the gambler must accept C , so the continuation value is $\mathbb{E}[C] = 1$. At stage 2, the support of B is contained in $[0, 1]$, so $B \leq 1$ always; since no realization of B strictly exceeds the threshold of 1, the gambler always passes to stage 3. (At $B = 1$ the gambler is indifferent; either action yields the same expected payoff, so we adopt the convention of rejecting ties.) The continuation value at stage 1 is therefore still 1, and since $A \leq 1$ always, the gambler again passes. Hence $G_{ABC} = \mathbb{E}[C] = 1$.
- **Order ACB .** At stage 3 the gambler must accept B , so the continuation value is $\mathbb{E}[B] = p$. At stage 2, both realizations of C (namely 0 and 1) satisfy $C \geq p$ when $p \leq 1$; in particular $C = 1 > p$ always, so the gambler always accepts C . The continuation value at stage 1 is therefore $\mathbb{E}[C] = 1$, and by the same argument as above $G_{ACB} = 1$.
- **Order CAB .** At stage 3 the gambler must accept B , so the continuation value is $\mathbb{E}[B] = p$. At stage 2, the gambler accepts $A = 1$ (since $1 > p$) and rejects $A = 0$ (since $0 < p$), giving the stage-2 continuation value

$$G_{AB} = p \cdot 1 + (1 - p) \cdot p = 2p - p^2 = \frac{1}{2},$$

where the last equality uses $p = \frac{1}{2}$. At stage 1, the threshold is $G_{AB} = \frac{1}{2}$; since $C = 1 > \frac{1}{2}$ always, the gambler accepts C immediately. Hence $G_{CAB} = \mathbb{E}[C] = 1$.

Prophet's value and ratio. The maximum $\max(A, B, C)$ takes values $\frac{1}{2}$, 1, and $\frac{1}{2\varepsilon}$ with probabilities $(1 - \varepsilon)(1 - p)^2$, $(1 - \varepsilon)(2p - p^2)$, and ε , respectively, so

$$\mathcal{P} = \frac{1}{2\varepsilon} \cdot \varepsilon + 1 \cdot (1 - \varepsilon)(2p - p^2) + \frac{1}{2} \cdot (1 - \varepsilon)(1 - p)^2 = \frac{1}{2} + \frac{3}{4}(1 - \varepsilon) \xrightarrow{\varepsilon \rightarrow 0} \frac{5}{4}.$$

Since A and B are identically distributed, swapping them does not change the gambler's value, so it suffices to consider the three orderings ABC , ACB , and CAB . Backward induction on each (Section 4.1.6) gives $G_{ABC} = G_{ACB} = G_{CAB} = 1$ for $\varepsilon = 0$. Therefore,

$$\frac{\mathcal{P}}{\mathcal{G}} \xrightarrow{\varepsilon \rightarrow 0} \frac{5}{4}.$$

4.2 Three Distributions: Unequal Permutations

In the equal-permutations setting the gambler obtains the same value from every ordering. When this symmetry is dropped, the gambler chooses the best ordering from a fixed set. We call such a set *improvable* if adding a permutation strictly increases the gambler's value.

Conjecture 4.5 (Beyhaghi [Bey19]). *A non-improvable set of maximal permutations contains at least three permutations, with each distribution appearing first in at least one of them.*

Each permutation in which X_i appears first determines the rightmost support of X_i : the gambler sets a threshold equal to the continuation value of the remaining two-variable subproblem. The two possible orderings of the other two distributions yield eight combinatorially distinct cases (Table 4.1).

Case	Permutation set	A_{sup}	B_{sup}	C_{sup}
1	$\{ABC, BCA, CAB\}$	BC	CA	AB
2	$\{ABC, BCA, CBA\}$	BC	CA	BA
3	$\{ABC, BAC, CAB\}$	BC	AC	AB
4	$\{ABC, BAC, CBA\}$	BC	AC	BA
5	$\{ACB, BCA, CAB\}$	CB	CA	AB
6	$\{ACB, BCA, CBA\}$	CB	CA	BA
7	$\{ACB, BAC, CAB\}$	CB	AC	AB
8	$\{ACB, BAC, CBA\}$	CB	AC	BA

Table 4.1: Eight non-improvable cases. Case 3 ($A_{\text{sup}} = BC$, $B_{\text{sup}} = AC$, $C_{\text{sup}} = AB$) recovers the equal-permutations support structure.

Optimization setup. The offline optimum is computed as in previous sections, with support boundaries adjusted per case. The normalization $A_4 + BC = B_4 + CA = C_4 + AB = 1$ fixes the online optimum to 1. Two subcases arise depending on whether $C_{\text{sup}} \leq C$ (subcase 2) or $C \leq C_{\text{sup}}$ (subcase 1), for sixteen problems in total, each solved via SLSQP with 1,000 random restarts.

Numerical results. Subcase 2 runs consistently reach ratios within 5×10^{-4} of $\frac{5}{4}$, with cases 3 and 4 converging to within 3×10^{-7} . Subcase 1 results should be treated as preliminary: the rerun of case 1 with the corrected A_{sup} boundary shows a normalization residual of ≈ 0.04 , indicating the constraint is not cleanly enforced and the subcase 1 figures may not be fully reliable.

Structure of the converged solutions. The reliable subcase 2 results reveal two structural families, both consistent with ratio $\frac{5}{4}$. **Family 1** (cases 1, 2): $A \approx B \approx 1 - 1/\sqrt{2} \approx 0.293$, $C \approx 1$, $C_{\text{sup}} \approx \frac{1}{2}$, matching the conjectured tight example. **Family 2** (cases 3, 4): $A \approx 0$, $B \approx \frac{1}{2}$, $C \approx 1$, $C_{\text{sup}} \approx \frac{1}{2}$, consistent with the degenerate embedding where $X_A \rightarrow 0$ and the problem reduces to the two-distribution worst case.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis studies the free-order prophet inequality when the identical-distributions assumption is dropped. For two distributions, we prove a tight worst-case competitive ratio of 0.80, strictly above the $\frac{1}{2}$ adversarial-order guarantee. For three distributions, structural reductions narrow the space of worst-case instances and support the conjecture that the tight ratio remains 0.80, though a full proof is open. Taken together, the results identify the exact answer for $n = 2$ and develop machinery that appears extensible to larger instances.

5.2 Future Work

Completing the $n = 3$ bound. The equal-permutations reductions reduce the problem to ruling out a remaining family of candidate distributions. A sharper global argument here would likely also clarify the setting for arbitrary n .

Multiple local optima. The numerical experiments suggest the $\frac{5}{4}$ ratio is achieved by structurally distinct instance families. In the two-distribution case, the tight example has A concentrating near $\frac{1}{2}$ with a rare large outcome and $B \sim \text{Bernoulli}(\frac{1}{2})$. The conjectured three-distribution tight example reverses the roles: $A = B \sim \text{Bernoulli}(1 - \frac{1}{\sqrt{2}})$ and C concentrates near $\frac{1}{2}$ with a rare large outcome. A third family embeds the two-distribution worst case directly: add a degenerate $D \equiv \mu_D$, place it last, and the gambler reduces to the two-variable subproblem recovering ratio $\frac{5}{4}$. Whether these families are connected, and how they generalize to $n > 3$, is an open structural question.

General n . The exact worst-case ratio for general n is open. The non-*iid* ratio is bounded above by the *iid* ratio since the *iid* setting is a special case. The evidence here suggests the ratio may hold at 0.80 through $n = 3$, but begin decreasing for $n \geq 4$ and potentially stabilize near the known lower bound of 0.7258 [BC23] for $n \geq 5$. Identifying where the drop begins would clarify how distributional heterogeneity affects the free-order problem at scale.

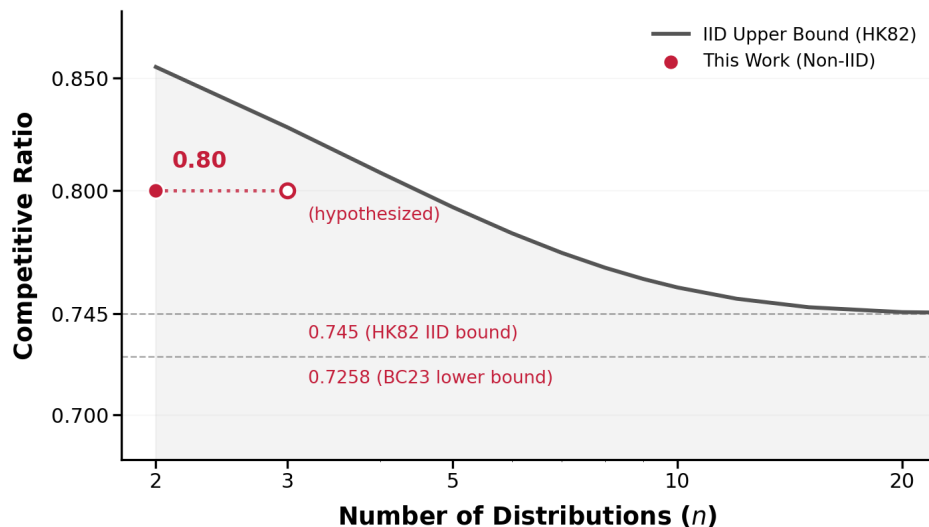


Figure 5.1: Schematic summary of how the competitive ratio behaves as the number of distinct non-*iid* distributions increases.

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A Area Characterization of Optimal Gambler Thresholds for the *iid* case

This appendix develops closed-form expressions for the optimal gambler thresholds V_i (as defined in [HK82]) in terms of areas under the survival function. We first recall the necessary setup, then state and prove the two main results.

Setup and Definitions

Definition .1 (Threshold intervals and areas). For optimal gambler thresholds $0 = V_0 < V_1 < \dots < V_n$, define the intervals

$$I_i = [V_i, V_{i+1}), \quad i = 0, \dots, n-1, \quad \text{and} \quad I_n = [V_n, \infty).$$

Define the corresponding areas under the survival function on each interval,

$$A_i = \int_{I_i} (1 - F_X(x)) dx, \quad i = 0, \dots, n,$$

and let $A = \sum_{i=0}^n A_i$.

The area decomposition $\mathbb{E}[X] = A$ follows immediately by partitioning the integral $\mathbb{E}[X] = \int_0^\infty (1 - F_X(x)) dx$ over the intervals $\{I_i\}$.

Definition .2 (Truncated variables). For fixed thresholds $0 = V_0 < V_1 < \dots < V_n$, define

$$X_i := \max(X, V_i), \quad i = 1, \dots, n.$$

Main Results

Lemma .3 (Threshold recurrence via areas). *For all $i \in \{1, \dots, n-1\}$,*

$$V_{i+1} = V_i + A - \sum_{k=0}^{i-1} A_k.$$

Proof. By the backwards-induction characterization of the gambler's optimal value, $V_{i+1} = \mathbb{E}[X_i] = \mathbb{E}[\max(X, V_i)]$. Applying the survival-function identity $\mathbb{E}[Z] = \int_0^\infty (1 - F_Z(x)) dx$ to $Z = X_i$, and noting that $F_{X_i}(x) = 0$ for $x < V_i$ and $F_{X_i}(x) = F_X(x)$ for $x \geq V_i$,

$$\mathbb{E}[X_i] = \int_0^{V_i} 1 dx + \int_{V_i}^\infty (1 - F_X(x)) dx = V_i + A - \int_0^{V_i} (1 - F_X(x)) dx.$$

Partitioning $[0, V_i)$ into the intervals $I_0, \dots, I_{i-1}$ (Definition .1),

$$\int_0^{V_i} (1 - F_X(x)) dx = \sum_{k=0}^{i-1} A_k,$$

which gives the stated recurrence. □

Theorem .4 (Closed form for optimal thresholds). *For all integers $1 \leq i \leq n$,*

$$V_i = iA - \sum_{k=0}^{i-2} (i - k - 1) A_k.$$

(The sum is empty and equals zero when $i = 1$.)

Proof. We proceed by induction on i .

Base case ($i = 1$). By the area decomposition $\mathbb{E}[X] = A$ and the base threshold $V_1 = \mathbb{E}[X]$ (i.e. the gambler's value after one draw equals the expectation), we have $V_1 = A$, matching the formula with an empty sum.

Inductive step. Assume the formula holds for some $i = j$:

$$V_j = jA - \sum_{k=0}^{j-2} (j - k - 1) A_k.$$

Applying Lemma .3,

$$V_{j+1} = V_j + A - \sum_{k=0}^{j-1} A_k = (j+1)A - \sum_{k=0}^{j-2} (j - k - 1) A_k - \sum_{k=0}^{j-1} A_k.$$

Combining the two sums over the shared index range $k = 0, \dots, j-2$,

$$(j - k - 1) + 1 = (j + 1) - k - 1,$$

and the remaining $k = j-1$ term contributes A_{j-1} from the second sum alone. Thus,

$$V_{j+1} = (j+1)A - \sum_{k=0}^{j-1} ((j+1) - k - 1) A_k,$$

which is exactly the formula for $i = j+1$, completing the induction. □